

Do ESG negative emotions on social media improve ESG disclosure quality? A study based on machine learning methods

Received: 9 April 2025

Accepted: 9 July 2026

Published online: 23 July 2026

Cite this article as: Wang, W. Do ESG negative emotions on social media improve ESG disclosure quality? A study based on machine learning methods. *Humanit Soc Sci Commun* (2026). <https://doi.org/10.1057/s41599-026-08360-z>

Wanting Wang

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

Do ESG negative emotions on social media improve ESG disclosure quality? A study based on machine learning methods

Abstract: ESG information has become a focus of attention for stakeholders. The study takes Chinese A-share listed companies from 2011 to 2021 as the research sample and employs the Word2vec algorithm and BERT model from machine learning methods to measure ESG negative emotions on social media (ENESM) based on the number of negative posts on investor interaction platforms. It finds that ENESM can compel companies to improve their ESG disclosure quality (EDQ). Mechanism testing shows that ENESM lead to a decrease in analyst attention, an increase in financing costs, and a decrease in customer stability for companies, forcing them to improve the EDQ. The more long-term institutional investors hold shares and the lower internal control quality, the stronger the positive impact of ENESM on the EDQ. Further research has found that compared to E (environmental) information and S (social) information, the positive correlation between ENESM and G (governance) disclosure quality is stronger. Although ENESM lead to companies with medium EDQ turning ESG greenwashing, strong regulation can effectively constrain this behavior. Non-ESG negative emotions on social media also drives companies to improve EDQ. ENESM further leads to analyst forecast dispersion and error. Moreover, increased financing constraints and bankruptcy risk compel companies to invest more in ESG disclosure costs in response to ENESM. The research has expanded the relevant literature on social media and EDQ, providing useful references for companies to cope with negative public opinion and improve EDQ.

Keywords: ESG negative emotions on social media; ESG disclosure quality; Analyst attention; Financing costs; Customer stability

JEL:M14

1. INTRODUCTION

The rapid economic development and the growing information demands of market participants are driving various investors to interact through social media (Wang et al., 2020)^[1]. On the background of increasingly prominent environmental issues, investors began to emphasize the ESG of enterprises (Li et al., 2022)^[2]. ESG has become a hot topic of discussion on social media platforms (Li et al., 2025)^[3]. However, investors' emotions on social media have different tendencies, among which negative emotions will accelerate aggregation and trigger problems. In July 2023, German media exposed that products sold to Europe by Deye Co., Ltd. were inconsistent with their VDE certification declarations. The news spread rapidly on Twitter and YouTube, causing the company's stock price to plummet by over 40% within just two months. The digital economy has optimized the interaction ways between enterprises and investors. However, it has also intensified the spread and diffusion of ESG negative emotions on social media (ENESM). Therefore, it is necessary to examine the impacts of ENESM and explore corresponding countermeasures.

In fact, enterprises can adjust their decisions promptly to avoid the fermentation of negative emotions on social media (Dyck et al., 2008)^[4]. They can respond and clarify through social media platforms (Lee et al., 2015)^[5]. For example, in 2015, Dongguan Securities issued a clarification announcement in response to the negative information about "Dongguan Securities' risk warning for 50 million VIPs" in the stock bar. They can also adjust their development strategies to response, such as technological innovation (Yang & Zhang, 2021)^[6] and changing merger and acquisition

plans (Luo & Dou, 2022)^[7]. Moreover, they can improve information disclosure behaviors, such as increasing the voluntary disclosure of earnings forecasts (Wang et al., 2020)^[8]. However, research about information disclosure has not received sufficient attention, and it is necessary to deeply explore the information disclosure behaviors of enterprises under the ENESM. ENESM refers to the dissatisfaction and skepticism expressed by investors on social platforms regarding corporate ESG practices. The question then arises: when faced with ENESM, will companies respond by enhancing their ESG disclosure?

In addition, existing research shows that ESG disclosure can serve as a reputation management tool to help enterprises gain the trust of external stakeholders (Xie et al., 2019)^[9], reduce enterprise risks (Reber et al., 2022)^[10], and experience lower idiosyncratic volatility, better liquidity, and higher institutional ownership (Chowdhury et al., 2021)^[11]. Furthermore, research indicates that ESG can serve as an effective risk management tool for enterprises. Specifically, CSR initiatives have been shown to help companies mitigate the deterioration of market quality following the outbreak of war (Shang & Fu, 2024)^[12]. ESG practices can assist firms in addressing the negative impact of investors' climate change sentiments on corporate value (Bagh et al., 2025)^[13]. Additionally, companies with higher systemic risk tend to increasingly utilize ESG strategies to cope with climate policy uncertainty (Dai & Jiang, 2025)^[14]. In reality, the awareness of ESG disclosure among listed enterprises is also constantly strengthening. In China, the number of listed companies disclosing ESG reports exceeded 1,000 in 2020^①, and increased to 2,481 by 2024 with a disclosure rate reaching 46.09%^②. The above analysis shows that it is meaningful to analyze the impact of the ENESM on the EDQ, which is helpful to safeguard investor rights and improve the EDQ.

However, there is a game process in whether enterprises choose to improve the EDQ. According to stakeholder theory, enterprises fulfill social responsibilities for their stakeholders (Qureshi et al., 2020)^[15], utilizing ESG disclosure to enhance corporate reputation (Zhou et al., 2020)^[16], thereby generating a governance effect on the organization (Li et al., 2022)^[2]. But, information cost theory posits that companies, in order to safeguard profits and shareholder wealth, are often reluctant to incur substantial costs associated with ESG disclosure (Hermalin & Weisbach, 2012^[17]; Lin et al., 2019^[18]; Wang et al., 2022^[19]).

So, how exactly does ENESM affect EDQ? To address this question, this study takes Chinese A-share listed companies from 2011 to 2021 as the sample, employs machine learning methods along with the BERT model to measure ENESM, and constructs the regression model to examine the relationship between ENESM and EDQ.

Compared with previous literature, the article has the following contributions:

First, this study expands the relevant research on enterprises' responses to the ENESM from the perspective of ESG disclosure. Previous literature has examined coping strategies such as technological innovation (Yang & Zhang, 2021)^[6], adjusting executive compensation (Dou & Luo, 2020)^[20], changing merger and acquisition preferences (Luo & Dou, 2022)^[7], and disclosing key audit matters (Wu et al., 2022)^[21]. However, the current ESG concept prevails, and improving EDQ is an important way to win the trust of investors. This paper confirms that improving EDQ is an effective measure to deal with ENESM, which expands the research on EDQ and ENESM.

Second, this study comprehensively employs information cost theory and stakeholder theory to conduct an in-depth analysis of the dynamic trade-off process through which ENESM compels enterprises to enhance EDQ. The research reveals that ENESM leads to reduced analyst coverage, increased financing costs, and weakened customer stability for enterprises. These adverse effects drive companies to reassess the costs of ESG disclosure against the

① The data is sourced from the "2024 China ESG Development White Paper".

② The data is sourced from the "Analysis Report on Sustainability Information Disclosure of Listed Companies 2025" compiled by the China Association of Public Companies.

economic benefits of improving EDQ, ultimately prompting them to elevate EDQ as a response to ENESM. This breakthrough transcends the traditional paradigm that positions information cost theory and stakeholder theory as opposing frameworks in previous research. Furthermore, this paper investigates the hedging effect of EDQ against ENESM, providing specific implementation pathways for enterprises to utilize ESG disclosure in preventing and mitigating ENESM risks.

Thirdly, this paper deepens the research on the relationship between ENESM and EDQ. The study examined the heterogeneous effects of institutional investors and internal controls, elucidating the preference of companies for environmental information when disclosing ESG information. At the same time, it was found that low EDQ can trigger ESG greenwashing, but it can be effectively curbed in a strong regulatory environment. By distinguishing the themes of social media negative sentiment, it is found that companies use ESG disclosure as a strategic tool to address non-ESG negative sentiment on social media. Meanwhile, beyond the quantity of AA, ENESM further leads to analyst forecast dispersion and error. Moreover, increased financing constraints and bankruptcy risk compel companies to invest more in ESG disclosure costs in response to ENESM. This article provides a personalized ESG disclosure response strategy for enterprises, and finds an optimization path for the rational use of ESG disclosure.

2. HYPOTHESES DEVELOPMENT

2.1 Adverse Consequences of ENESM

This study argues that ENESM poses serious challenges to corporate sustainable development by imposing triple pressures through the "information channel, capital channel, and market channel," which reduce analyst coverage, increase financing costs, and lower customer stability for enterprises.

From the information channel perspective, this paper argues that ENESM leads to a reduction in analyst attention (AA) for companies. Securities analysts are crucial intermediaries of information in the capital market (Liu & Li, 2018)^[22]. However, AA is limited (Dellavigna & Pollet, 2009)^[23] and is often directed towards companies that are highly anticipated or viewed optimistically by the market (Zhang & Wu, 2021)^[24], resulting in the "selective neglect" of firms with significant ENESM. This occurs for several reasons: firstly, ESG-related non-financial information often lacks a standardized assessment framework (EFRAG, 2021)^[25]. When ENESM emerges, analysts face significantly increased information processing costs. Simultaneously, stocks affected by negative public opinion are prone to decreased liquidity. These factors lead to a diminishing return on analysts' research efforts. Second, when dealing with ENESM, analysts face unique risks: on one hand, publishing pessimistic research reports may permanently damage their relationship with the company's management, resulting in the loss of valuable research channels; on the other hand, analysts need to protect their own reputation (Liu & Li, 2018)^[22]. Making incorrect predictions based on uncertain ENESM can directly harm their personal credibility. "Selective neglect" becomes a more prudent strategy. When AA decreases significantly, a company's visibility and credibility in the capital market are simultaneously undermined.

From the capital channel perspective, this paper argues that ENESM leads to increased financing costs (FC) for enterprises. Access to financing is a critical foundation for subsequent corporate development. Investors can obtain comprehensive information about companies through social media (Zheng et al., 2022)^[26] and strive to avoid investment losses caused by hidden bad news. ENESM raises investor concerns about the target company's production operations and cash reserves, resulting in a significant increase in its FC. On one hand, when a company is involved in ESG negative incidents, the market interprets this as indicating serious governance deficiencies and potential risks

within the company (Asante-Appiah & Lambert, 2022)^[27]. To protect their own interests, investors demand higher risk compensation (La & Bernini, 2022)^[28]. On the other hand, institutional investors adhering to responsible investment principles tend to favor companies with more positive ESG publicity (He et al., 2025)^[29], substantially reducing potential capital sources for companies with significant ENESM. Therefore, ENESM not only directly increases FC by raising risk premiums but also indirectly exacerbates corporate capital constraints by restricting financing channels, posing severe challenges to their financial health and long-term value.

From the market channel perspective, this paper argues that ENESM leads to weakened customer stability (CS) for enterprises. Companies are closely interconnected through supply chains (Zeng & Tang, 2023)^[30], and customers are the source of corporate cash flow. A supplier's ENESM significantly undermines its CS. Due to the risk transmission effect within supply chains (Xiao et al., 2024)^[31], a supplier's ENESM is no longer merely its own problem but directly transmits risks to upstream or downstream partners. When a supplier is involved in an ESG scandal, its supply chain customers face substantial external pressure. These customers' own brand reputation, investor relations, and ESG ratings may be adversely affected by their association with the supplier's ENESM. Customers are motivated to avoid risks and may initiate rigorous reviews of suppliers with ENESM, potentially suspending cooperation, reducing orders, or even removing them from qualified supplier lists. Consequently, ENESM is no longer just a crisis of brand image but, through supply chain dependencies, leads to the loss of critical orders and stable customers.

2.2 Impact Mechanism of ENESM on EDQ

The Information Cost Theory and the Stakeholder Theory are the two major theories in current ESG information disclosure research. The Information Cost Theory holds that ESG disclosure requires enterprises to bear relatively high costs (Wang et al., 2022)^[19]. These costs include various equipment investments, management and evaluation, personnel recruitment and the opportunity cost of funds, etc. (Wang et al., 2022)^[19]. After incurring the aforementioned costs to improve the EDQ, enterprises may experience reduced profits and diminished shareholder wealth (Lin et al., 2019)^[18]. This often leaves enterprises lacking the motivation for voluntary disclosure under a short-term profit orientation, especially tending to engage in selective disclosure or passive responses in the absence of mandatory constraints. Stakeholder Theory posits that enterprises will enhance the EDQ to maintain relationships with their stakeholders (Freeman, 1984)^[32]. This can prompt external stakeholders to form more positive judgments about the enterprise (Xie et al., 2019)^[9], help enhance corporate value (Raimo et al., 2021)^[33], and reduce corporate risks (Rember et al., 2022)^[10]. Most existing studies unilaterally support either the Information Cost Theory or Stakeholder Theory, viewing them as mutually opposing. However, this paper argues that when enterprises face ENESM, various stakeholders of the enterprises will re-evaluate their sustainable development capabilities and governance levels, and exert pressure on the enterprises through specific behaviors in information channels, capital channels, and market channels. Management will weigh the costs of information disclosure against stakeholder demands, and ultimately respond to ENESM by improving EDQ.

Then, within the synergistic theoretical framework of the Information Cost Theory and Stakeholder Theory, how will enterprises balance the costs of information disclosure and the interests of stakeholders to address the aforementioned pressures? This paper argues that under this theoretical framework, the impact of ENESM on EDQ presents a dynamic evolutionary process. When enterprises are not confronted with ENESM, the Information Cost Theory is more conducive to explaining the enterprise's behavior: as the collection, verification, and disclosure of ESG information require substantial resources and may expose commercially sensitive information, enterprises

naturally tend to minimize disclosure to save costs (Wang et al., 2022)^[19]. However, when ENESM erupts and leads to reduced AA, increased FC, and weakened CS, the implicit costs of maintaining persistently low EDQ far exceed the explicit costs required to improve EDQ. When ENESM breaks out, stakeholders perceive NESM as a signal of governance failure and express their collective demands through actions such as capital withdrawal, tighter financing conditions, and order cancellations. At this juncture, the enterprise's cost structure undergoes a fundamental restructuring: the implicit costs of maintaining low-level disclosure (e.g., soaring financing premiums, losses from customer churn) begin to significantly surpass the explicit costs of enhancing information disclosure. This cost inversion phenomenon prompts enterprises to invest in EDQ, aiming to avoid greater economic losses by conveying more transparent and higher-quality EDQ to external stakeholders. This paper will elaborate on the enterprise's strategic adjustment process under ENESM from three perspectives: analyst attention under information pressure, financing costs under capital pressure, and customer stability under market pressure.

First, a reduction in AA will compel companies to enhance their EDQ in response to NESM. Analysts primarily exert influence on management through analytical reports, forecasts, and other means (Tan et al., 2016)^[34]. By interpreting media information and publishing reports, they can amplify the impact of negative media narratives (Yu et al., 2011)^[35]. A reduction in AA directly sends a strong risk signal to a broader range of stakeholders, undermining the company's market credibility and legitimacy. This leads to increased pressure from narrowed financing channels and rising capital costs (Li et al., 2023)^[36]. To maintain relationships with these core stakeholders and regain market trust, companies urgently need to establish new, credible communication channels to restore market confidence. ESG disclosure can serve as an impression management tool, preserving and enhancing the company's reputation among external stakeholders. This, in turn, enables external stakeholders, such as analysts, to form more positive assessments of the company's value and operational performance (Xie et al., 2019)^[9]. Consequently, pressure from the capital market will drive companies to no longer view ESG disclosure as a burden but as a strategic imperative.

Secondly, the increase in FC will force enterprises to gain market confidence through ESG information disclosure. Studies by Barry & Brown (1984)^[37] and Handa & Linn (1993)^[38] have theoretically demonstrated that investors tend to assign a higher risk level to enterprises with low information disclosure. Consequently, they show lower demand for such stocks, offer lower prices, and require higher investment returns, ultimately leading to higher FC for the companies. To rebuild trust relationships with key stakeholders such as capital providers, enterprises must enhance the EDQ to reduce their risk premium. Existing studies have shown that the long-term efforts made by enterprises through activities such as ESG can be better perceived by investors, which can reduce FC (Raimo et al., 2021^[33]; Dahiya & Singh, 2020^[39]; Li et al., 2022^[40]). Corporate reputation enhancement can facilitate its relationships with financial institutions and investors, thereby securing more financial support (Orlitzky et al., 2003^[41]). In such cases, companies are willing to incur ESG information disclosure costs in order to secure sustained financing channels.

Finally, the weakening of CS can prompt enterprises to confront the survival pressure in the product market directly and implement strategic self-rescue by improving the EDQ. As the substantive impacts such as order loss and the decline in contract renewal rates become apparent, the static balance described by the Information Cost Theory has been disrupted: the opportunity cost of maintaining the original disclosure level (i.e., the revenue loss caused by customer churn) has far exceeded the direct investment of enhancing EDQ. At this point, the dynamic game revealed by the Stakeholder Theory begins to dominate corporate decision-making. Atif & Ali (2021)^[42] found that high EDQ can enhance customer satisfaction and loyalty, ensure the revenue stability and cash flow during crises. Higher CS indicates that the enterprise's operating conditions are stable and its future development prospects are good, reducing their expectations of operating risks (Li et al., 2023)^[43]. The weakening of CS not only causes pressure on the current

performance of enterprises, but also conveys a strong signal to the capital market that the sustainable development capacity of enterprises has been impaired. To break this chain reaction, enterprises must rebuild customer trust by improving the quality of ESG information disclosure.

In summary, within the synergistic framework of Information Cost Theory and Stakeholder Theory, although ENESM leads to reduced AA, increased FC, and weakened CS for enterprises, companies ultimately choose to enhance EDQ in response to ENESM by weighing the costs of ESG disclosure against the implicit costs arising from adverse consequences, thereby addressing stakeholder concerns. Fig.1 illustrates the theoretical framework of the paper. Accordingly, we propose the following hypothesis:

H1a: ENESM are positively correlated with the EDQ.

H1b: ENESM leads to reduced AA, increased FC, and weakened CS for enterprises, thereby compelling them to enhance EDQ.

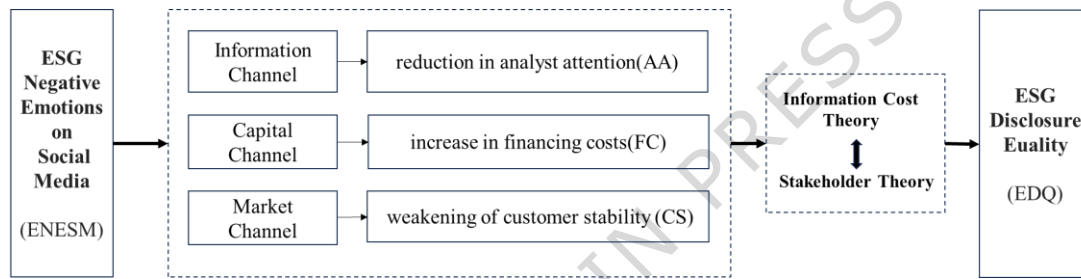


Figure 1. Theoretical Framework Diagram

3. RESEARCH DESIGN

3.1 Data and Sample

This paper takes A-share listed companies in 2011-2021 as the research object, and excludes the data missing and ST samples, and finally obtains 7628 company annual observations. ESG data is obtained from Bloomberg ESG evaluation database, and financial data is obtained from CSMAR database. The negative ESG-themed investor interaction texts used in this study are sourced from the Shenzhen Stock Exchange's Interactive Easy platform and the Shanghai Stock Exchange's SSE e-Interaction platform. All continuous variables are winsorized at the upper and lower 1% levels.

3.2 Dependent Variable

Referring to Wang et al. (2022)^[19], we use Bloomberg ESG scores to measure the quality of corporate ESG information disclosure. The Bloomberg ESG score is reflected in specific numerical values and has a total of 120 measurement indicators, which can comprehensively measure the EDQ. Therefore, this paper measures the EDQ by taking the logarithm of Bloomberg ESG score +1.

3.3 Independent Variable

Referring to Huang & Jiang (2022)^[44], we added the logarithm of the number of negative posts+1 in the Investor Interaction Platform on the Shenzhen Stock Exchange (SZSE) and Shanghai Stock Exchange (SHSE) to measure ENESM. Referring to Zhang et al (2025)^[45], the rationale for selecting investor questions on exchange platforms as a

proxy variable for the sentiment of retail investors from stock message boards is as follows: To establish direct and standardized communication between investors and listed companies while protecting the legitimate rights and interests of minority shareholders, the Shenzhen Stock Exchange and Shanghai Stock Exchange have successively launched the "Shenzhen Exchange Interactive Easy" and "SSE E-Interaction" platforms. In China, other similar social media platforms include investor online forums and unofficial social media channels, but each has its limitations. On online forums such as GUBA and Snowball forums, users cannot communicate directly with companies. On unofficial social media platforms like Sina Weibo and WeChat Official Accounts, content is typically initiated by companies or professionals and lacks mandatory regulatory requirements. Consequently, these platforms suffer from poor interactivity and are inadequate as primary channels for investors to express their concerns. In contrast, on the exchanges' investor interaction platforms, investors can log in to submit questions to listed companies, which are required to designate management personnel to provide timely responses. All interactions are publicly accessible to website visitors and are incorporated into the regulatory assessment of information disclosure.

Meanwhile, referring to Li et al. (2025)^[3], the word2vec method and BERT model from machine learning are employed to screen negative investor posts. The specific steps are as follows:

(1) Construction of an ESG Dictionary

First, nine major policy documents covering the environmental, social, and governance dimensions were selected from the Shanghai and Shenzhen Stock Exchanges. After word segmentation using Python software and manual merging and screening, 174 ESG keywords were obtained. Second, based on Q&A interactions from all sample enterprises, the word2vec machine learning method was applied to extract similar terms for the ESG keywords, ultimately resulting in 279 ESG-themed keywords.

(2) Prediction of Sentiment Orientation

First, the ESG dictionary is matched with information conveyed by investors on two types of platforms to determine whether the 279 keywords appear in investor questions. Second, as the BERT model in deep learning demonstrates significant advantages in recognizing contextual semantics and handling complex topics, we leverage pre-trained model parameters and manually labeled tone texts to conduct pre-training and fine-tuning using Python, ultimately classifying the sentiment of interactive texts. At the same time, we examined the applicability of the BERT model. The F1 score is 0.7806, and the confusion matrix of the BERT model based on all posts is shown in Figure 2^③. Finally, the ESG-related negative questions raised by each investor for each enterprise are aggregated to obtain the total volume of negative ESG questions from investors. After adding 1 and taking the natural logarithm, the annual intensity of negative ESG questions from investors (ENESM) is derived.

③ To further test the robustness of the BERT model in predicting social media sentiment, this study draws a subsample of 1,000 posts to assess model validity. The resulting two-human-annotator κ is 0.9026, with a precision of 0.7794, recall of 0.6242, and F1 score of 0.6634.

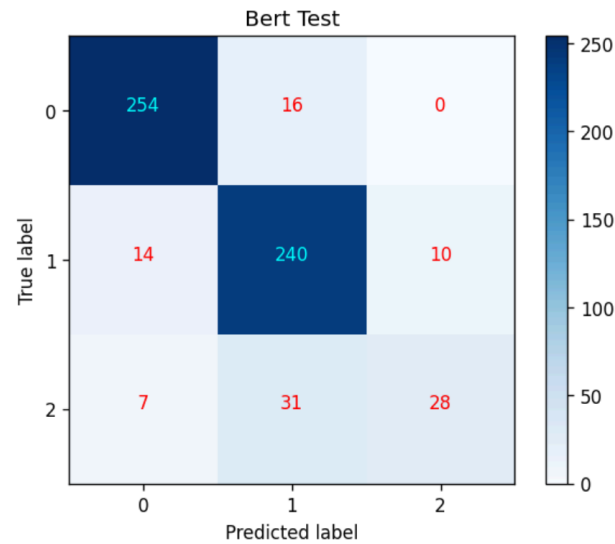


Figure 2. Confusion Matrix of the BERT model based on all posts

3.4 Mechanism Variables

Analyst attention (AA) is calculated using the number of research reports. Financing cost (FC) is calculated based on the MPEG model(Cai et al.,2022)^[46].Customer stability (CS) is calculated as the ratio of the number of the top five customers reported in the previous fiscal year to 5(Pan and Zhang, 2020)^[47].

3.5 Control Variables

Referring to Wang et al. (2022)^[19], the following control variables are selected: Asset-liability ratio (LEV), Executive compensation (PAY), Return on net assets (ROA), Board size (BOARD), Dual employment (DUAL), Price-earnings ratio (EPS), Revenue growth rate (GROWTH), Book-to-market ratio (MB), Ownership concentration (TOP5), Total assets (SIZE). In addition, firm fixed effects and year fixed effects are also controlled. The definitions of the variables are shown in Table 1.

TABLE 1 Variable definition.

Variables	Measures
ESG disclosure quality (EDQ)	Ln(Bloomberg ESG score +1)
ESG Negative emotions on social media (ENESM)	Ln(negative post number+1)
Analyst attention (AA)	Number of research reports
Financing cost(FC)	The equity capital cost
Customer stability (CS)	Number of the top five customers reported in the previous fiscal year /5
Asset-liability ratio (LEV)	Liabilities/total assets
Executive compensation (PAY)	Ln(compensation of the top three executives+1)
Return on net assets (ROA)	Net profit/ total assets
Board size (BOARD)	Ln(board member numbers +1)
Dual employment (DUAL)	1 if the chairman serves as the CEO,0 otherwise

Price-earnings ratio (EPS)	Net profit/total number of shares
Revenue growth rate (GROWTH)	Growth rate of operating revenue
Book-to-market ratio (MB)	Total assets/ market value
Ownership concentration (TOP5)	Shareholding ratio of the top five shareholders
Total assets (SIZE)	Ln(total assets+1)

3.5 Research model

To empirically examine Hypothesis H1, we establish the following fixed-effects regression model. The specification of Model (1) is presented below:

$$EDQ_{i,t} = \beta_0 + \beta_1 ENESM_{i,t} + \beta_2 \sum CONTROL S_{i,t} + FIRM_i + YEAR_t + \varepsilon_{i,t} \quad (1)$$

This paper mainly focuses on the coefficient of β_1 in model (1). Among them, $ENESM_{i,t}$ represents negative emotions on social media, $EDQ_{i,t}$ represents ESG disclosure quality, $CONTROL S_{i,t}$ represents control variables, $FIRM_i$ and $YEAR_t$ represents firm fixed effects and year fixed effects respectively.

4. EMPIRICAL RESULTS

4.1 Descriptive statistics

Table 2 presents descriptive statistics. The mean, maximum and minimum values of EDQ is 3.358, 3.993, and 2.622, and the standard deviation is 0.262, which indicates that there are great differences in the EDQ. The mean, maximum, minimum and standard deviation of ENESM are 0.808, 3.219, 0, and 0.808. This indicates that the number of negative comments fluctuated greatly among different enterprises during the observation period.

T A B L E 2 Descriptive statistics.

variables	N	Mean	SD	Min	Max
EDQ	7628	3.358	0.262	2.622	3.993
ENESM	7628	0.808	0.865	0	3.219
AA	6457	31.17	30.40	1	115
FC	5920	0.128	0.051	0.033	0.314
CS	4152	0.746	0.381	0	1
LEV	7628	0.460	0.200	0.050	0.933
PAY	7628	14.80	0.722	12.68	16.36
ROA	7628	0.048	0.065	-0.302	0.211
BOARD	7628	2.271	0.175	1.792	2.708
DUAL	7628	0.241	0.428	0	1
EPS	7628	0.578	0.753	-1.607	3.020
GROWTH	7628	0.189	0.421	-0.606	3.092
MB	7628	0.590	0.298	0.084	1.173
TOP5	7628	0.539	0.155	0.197	0.865
SIZE	7628	23.13	1.188	19.37	25.74

4.2 Baseline results

Table 3 reports the regression results of Model 1. Columns (1) only include the core explanatory variables, year fixed effects and firm individual fixed effects. Column (2) includes all variables. As shown in Table 3, the coefficients of ENESM are all positive significantly. It indicates that ENESM contribute to the improvement of EDQ, thus verifying H1a.

TABLE 3 Baseline results.

	EDQ (1)	EDQ (2)
ENESM	0.006** (2.231)	0.005** (2.021)
LEV		-0.085*** (-2.956)
PAY		0.016** (2.195)
ROA		-0.043 (-0.775)
BOARD		-0.009 (-0.424)
DUAL		-0.005 (-0.629)
EPS		0.009 (1.477)
GROWTH		-0.009** (-2.008)
MB		-0.052*** (-2.973)
TOP5		0.053 (1.229)
SIZE		0.038*** (4.862)
Constant	3.034*** (226.495)	2.014*** (11.397)
Year	√	√
Firm	√	√
Obs	7,628	7,628
R2_a	0.689	0.696

Notes: ***p < 0.01; **p < 0.05; *p < 0.1. t statistics are in parentheses

4.3 Mechanism analysis

4.3.1 The mechanism of ENESM on EDQ

According to the theoretical analysis, ENESM can promote companies to improve EDQ by reducing AA,

increasing FC and CS. To test the above mechanism, this paper constructs model (2)..

$$M_{i,t} = \beta_0 + \beta_1 NESM_{i,t} + \beta_2 \sum CONTROLS_{i,t} + FIRM_i + YEAR_t + \varepsilon_{i,t} \quad (2)$$

In model(2), $M_{i,t}$ represents the mechanism variables, which respectively include AA(analyst attention), FC (financial expenses), and CS (customer Stability). The remaining variables are consistent with model (1).

Table 4 reports the regression results of Model 2. In Column (1), the coefficient of ENESM is negative significantly. In Column (2), the coefficient of ENESM is positive significantly. In Column (3), the coefficient of ENESM is negative significantly. Thus, H1b is confirmed.

T A B L E 4 Regression results of ENESM on AA/FC/CS

	AA (1)	FC (2)	CS (3)
ENESM	-1.164*** (0.386)	0.002* (0.001)	-0.014* (0.008)
LEV	-0.077 (4.484)	0.004 (0.010)	-0.030 (0.090)
PAY	1.182 (1.054)	0.005** (0.002)	-0.011 (0.021)
ROA	31.920*** (10.775)	0.110*** (0.031)	0.312* (0.189)
BOARD	-0.834 (3.082)	0.010 (0.008)	-0.068 (0.078)
DUAL	0.737 (1.228)	-0.003 (0.002)	-0.009 (0.022)
EPS	5.936*** (0.947)	-0.001 (0.002)	-0.021 (0.017)
GROWTH	-1.655** (0.712)	0.000 (0.002)	-0.057*** (0.016)
MB	-62.986*** (2.814)	0.092*** (0.006)	0.108** (0.050)
TOP5	-15.494*** (5.762)	-0.000 (0.013)	-0.283*** (0.109)
SIZE	15.999*** (1.222)	-0.006** (0.003)	0.036* (0.019)
Constant	-305.406*** (27.114)	0.155** (0.066)	0.066 (0.446)
Year	√	√	√
Firm	√	√	√
Obs	6,457	5,920	4,152
R2_a	0.321	0.212	0.0685

Notes:***p < 0.01; **p < 0.05; *p < 0.1. t statistics are in parentheses

4.3.2 The mechanism of EDQ hedging against ENESM' Adverse Consequences

We examine whether improving the EDQ can offset above adverse effects. Referring to Kim et al. (2021)^[48], we use a two-stage model to test the hedging effect of improving EDQ. The two-stage model is specifically represented as Model (3) and Model (4).

$$\Delta EDQ_{i,t} = \beta_0 + \beta_1 \Delta NESM_{i,t} + \beta_2 \sum \text{CONTROLS}_{i,t} + \text{FIRM}_i + \text{YEAR}_t + \varepsilon_{i,t} \quad (3)$$

$$\Delta M_{i,t} = \beta_0 + \beta_1 \overline{\Delta EDQ_{i,t}} + \beta_2 \sum \Delta \text{CONTROLS}'_{i,t} + \text{FIRM}_i + \text{YEAR}_t + \varepsilon_{i,t} \quad (4)$$

Among them, the fitted value of $\Delta EDQ_{i,t}$ represents the real impact of ENESM on EDQ. $\Delta M_{i,t}$ is the difference value between $M_{i,t}$ and $M_{i,t-1}$, and it is represented by AA, FC, and CS. Table 5 reports the regression results of Model 3 and Model 4. The results indicate that improving the EDQ can effectively hedge against a decrease in AA and an increase in FC. In terms of CS, compared to ENESM's significant increase in CS, the insignificant correlation between EDQ and CS indicates that it can effectively mitigate the decrease in CS.

TABLE 5 Regression results of EDQ hedging against AA/FC/CS

	AA (1)	FC (2)	CS (3)
$\overline{\Delta EDQ_{i,t}}$	432.656*** (29.195)	-0.187*** (0.059)	0.173 (0.439)
ΔLEV	13.678*** (4.626)	-0.001 (0.010)	0.098 (0.087)
ΔPAY	-0.078 (0.938)	0.006*** (0.002)	-0.028 (0.022)
ΔROA	18.982** (8.512)	0.044* (0.026)	0.112 (0.152)
ΔBOARD	0.240 (3.025)	0.002 (0.007)	0.012 (0.075)
ΔDUAL	1.143 (0.993)	-0.005** (0.002)	-0.003 (0.019)
ΔEPS	1.218 (0.925)	-0.005** (0.002)	0.007 (0.018)
ΔGROWTH	-0.023 (0.584)	-0.002 (0.001)	-0.017 (0.012)
ΔMB	-7.614*** (2.636)	0.066*** (0.006)	0.124** (0.049)
ΔTOP5	-16.848** (7.043)	-0.011 (0.014)	-0.021 (0.125)
ΔSIZE	7.310*** (1.993)	-0.007* (0.004)	-0.136*** (0.036)
constant	-1,282.979*** (89.010)	0.721*** (0.179)	0.047 (1.341)
Year	√	√	√
Firm	√	√	√
Obs	4,901	4,503	3,476

R2_a 0.178 0.181 0.0576

Notes:***p < 0.01; **p < 0.05; *p < 0.1. t statistics are in parentheses

4.4 Robustness tests

4.4.1 Alternative Measure of Dependent Variables

Besides the investor Q&A platforms provided by stock exchanges, Stock Bar Forum is also an important online venue for investor interaction. Referring to Huang and Jiang (2022)^[44], we re-measured ENESM using negative posts from investors on the Stock Bar Forum. Calculate the ratios and average daily number of posts by distinguishing between natural days and trading days. Firstly, we employ the ratio of the number of negative posts to the total number of posts within natural days (NR) as a measure for ENESM. Secondly, we also employ the logarithm of the number of negative posts within trading day days plus one (NT) to measure ENESM. The regression result is shown in column (1) and (2) of Table 6. The coefficients of NR and NT are all positive significantly, indicating that previous research results are reliable.

4.3.2 Replacement of Regression Model

To address the problem of estimation bias caused by unobservable factors, we construct the change model to estimations again. Referring to Luo and Du(2018)^[49], we incorporate the differences in variables of model (1) into the regression. The regression result is shown in column (3) of Table 6. The coefficient of Δ ENESM is also positive significantly.

4.4.3 Alleviate Omitted Variables

There may be biases caused by omitted variables in the study. We further controlled macro variables including EPU(Economic Policy Uncertainty) and MSR(Management Shareholding Ratio) in model (1). The result is reported in columns (4) of Table 6. The coefficient of ENESM is also positive significantly.

4.4.4 Test of Nonlinear Relationship

A nonlinear relationship may also exist between ENESM and EDQ. Accordingly, we incorporated the squared term of ENESM into the regression of Model (1). As columns (5) of Table 6 shows, the coefficients of ENESM² and ENESM are all not significantly. This result indicates that there is no nonlinear relationship between ENESM and EDQ, thereby confirming the validity of the conclusions previously.

TABLE 6 Results of robustness tests^④

	EDQ Ratio	EDQ Daily Posting	EDQ Change model	EDQ Omitted Variables	EDQ Nonlinear Relationship
	(1)	(2)	(3)	(4)	(5)
NR	0.163** (2.076)				
NT		0.008** (2.129)			
Δ ENESM			0.005** (2.331)		

^④All the variables in the transformation model are difference terms. To save space, the English names of the control variables in the benchmark regression are used instead.

ENESM				0.005**	0.002
				(2.037)	(0.414)
ENESM ²					0.001
					(0.563)
LEV	-0.086***	-0.088***	-0.028	-0.092***	-0.085***
	(-2.983)	(-3.065)	(-1.020)	(-3.099)	(-2.946)
PAY	0.016**	0.015**	0.003	0.015**	0.016**
	(2.244)	(2.111)	(0.557)	(2.120)	(2.190)
ROA	-0.048	-0.039	-0.007	-0.049	-0.043
	(-0.871)	(-0.715)	(-0.155)	(-0.870)	(-0.780)
BOARD	-0.008	-0.008	-0.003	-0.007	-0.009
	(-0.379)	(-0.348)	(-0.144)	(-0.308)	(-0.421)
DUAL	-0.004	-0.005	-0.006	-0.005	-0.005
	(-0.587)	(-0.597)	(-0.921)	(-0.624)	(-0.640)
EPS	0.011*	0.009	0.001	0.010	0.009
	(1.725)	(1.518)	(0.085)	(1.512)	(1.487)
GROWTH	-0.009**	-0.008*	0.000	-0.009**	-0.009**
	(-1.978)	(-1.767)	(0.009)	(-2.028)	(-2.005)
MB	-0.049***	-0.039**	-0.006	-0.050***	-0.052***
	(-2.823)	(-2.194)	(-0.368)	(-2.772)	(-2.975)
TOP5	0.057	0.071	0.020	0.051	0.053
	(1.315)	(1.645)	(0.425)	(1.149)	(1.231)
SIZE	0.037***	0.035***	0.010	0.039***	0.038***
	(4.758)	(4.492)	(0.960)	(4.849)	(4.853)
EPU				0.393***	
				(27.746)	
MSR				0.018	
				(0.542)	
Constant	1.982***	2.025***	0.019	0.063	2.017***
	(11.080)	(11.419)	(1.294)	(0.373)	(11.410)
Year	√	√	√	√	√
Firm	√	√	√	√	√
Obs	7,628	7,550	5,771	7,376	7,628
R2_a	0.696	0.697	0.279	0.693	0.696

Notes:***p < 0.01; **p < 0.05; *p < 0.1. t statistics are in parentheses

4.5 Endogeneity test

4.5.1 Propensity Score Matching Method

To alleviate the errors in research results caused by sample selection, we adopt the Propensity Score Matching (PSM) method. Referring to the Hou and Ding (2015)^[50], we focus on samples with higher ENESM as the treat group, other samples are the control group. The covariates are the control variables in model (1). Then we employ radius

matching(caliper=0.05) for the treatment and control groups.As columns (1) of Table 7 shows,the results after PSM are still consistent with the previous research.

4.5.2 Regression with lagged term

This study aims to determine whether enterprises improve their EDQ to repair their reputation, or whether poor EDQ itself is the source of ENESM. Following Zhang et al. (2022)^[51],independent variable LLEDQ is constructed (which takes the value 1 when the EDQ of the previous period is poor, and 0 otherwise), with the current period's ENESM as the dependent variable. If the coefficient of the two is significant, it indicates that poor EDQ in the previous year affects the current period's ENESM. If it is not significant, it suggests that in the research sample of this study, the findings are more consistent with the explanation that "enterprises enhance EDQ in response to ENESM to repair reputational damage."As columns (2) of Table 7 shows, LLEDQ does not significantly affect ENESM, thereby indicating that companies respond to ENESM by enhancing EDQ to repair reputational damage.

4.5.3 Different-GMM test

To further clarify the causal relationship between ENESM and EDQ, this study introduces the one-period lagged EDQ (LEDQ) into the model and re-examines the hypothesis using the dynamic difference GMM method. The estimation results are presented in column (3) of Table 7. ENESM is significantly positive at the 5% level, the p-value of the AR(1) test is significant, the p-value of the AR(2) test is not significant, and the p-value of the Hansen test is 0.288, which is not significant, indicating that the difference GMM estimation method is highly applicable. The result confirms that after accounting the dynamic changes in EDQ, the driving effect of ENESM on EDQ remains robust.

4.5.4 Regression Discontinuity Design

The firms listed on the Shanghai and Shenzhen 300 constituent stock lists in this article can attract more investors' attention. Referring to the research of An et al. (2020)^[52], we conduct a Regression Discontinuity Design (RDD) test based on the Shanghai and Shenzhen 300 Index. We construct the model(5) for RDD test.

$$NESM_{i,t} = \alpha + \tau ifHS300_{i,t} + \sum_{j=1}^k \delta_j (MV_{i,t}^j)^N + \gamma_j Rank_{i,t} + \theta_j INST_{i,t} + \varepsilon_{i,t} \quad (5)$$

Among them, "ifHS300" is a dummy variable, which takes the value of 1 when the annual observation value of an enterprise belongs to the list of constituents of the Shanghai and Shenzhen 300 Index, and 0 otherwise; "MV" refers to the de-meant tradable market value of individual stocks at the end of December of that year; "ENESM" is the order of the polynomial selected to maximize the Bayesian Information Criterion (BIC). Through screening and comparison of BIC values, this paper finally determines that "ENESM" is 2; "Rank" is the ranking of companies by tradable market value in each observation year; "INST" represents the shareholding ratio of institutional investors. The results are reported in columns (4) and columns (5) of Table 7. The coefficient of $\widehat{ENESM}$ is positive significantly.

4.5.5 Instrumental Variable Approach

Referring to research of Lewbel (1997)^[53] and Xi Mingming et al. (2025)^[54], we employ the cubic terms of the deviations between BENESM and its mean values at both the industry and provincial levels (IV1 and IV2) as instrumental variables for the 2SLS estimation. The results are reported in columns (6) to columns (9) of Table 7. The coefficient of ENESM is also positive significantly. The test results for weak instruments and over-identification indicate that the selection of instrumental variables is relatively reasonable.The results demonstrate that the findings remain consistent after controlling for endogeneity.

4.5.6 Difference-in-Differences (DID)

We further employ a Difference-in-Differences (DID) model with fixed effects to mitigate endogeneity issues, selecting the establishment of investor interaction platforms as an exogenous shock. Given that the Shenzhen Stock

Exchange's "Interactive Easy" platform was launched in 2010, followed closely by the Shanghai Stock Exchange's "e-Interaction" platform in 2013, the emergence of these platforms provides an ideal context for applying the DID methodology. Therefore, we focuses on A-share listed companies in Shanghai and Shenzhen from 2011 to 2021, treating the launch of the "Interactive Easy" platform in 2013 as a key exogenous event to assess the actual impact of investor interaction on EDQ. The DID regression model constructed in this study is specified as follows:

$$EDQ_{i,t} = \gamma_0 + \gamma_1 Treat_{i,t} \times Post_{i,t} + \gamma_4 Controls_{i,t} + FIRM_i + YEAR_t + \varepsilon_{i,t} \quad (6)$$

Among them, Treat is a dummy variable that equals 1 for listed companies on the Shanghai Stock Exchange and 0 otherwise. Post is also a dummy variable, which takes the value of 1 for the year 2013 and onwards. We find the coefficient of Treat×Post is significantly positive in columns (10) of Table 7, consistent with previous tests.

TABLE 7 Results of endogeneity tests^⑤

	EDQ PSM (1)	ENESM RLT (2)	EDQ GMM (3)	ENESM RDD (4)	EDQ RDD (5)	NESM IV (6)	EDQ IV (7)	ENESM IV (8)	EDQ IV (9)	EDQ DID (10)
ENESM	0.005** (0.003)		0.005* (0.003)				0.008*** (0.003)		0.009*** (0.003)	
LLEDQ		0.018 (0.037)								
LEDQ			0.729*** (0.047)							
ifHS300				0.068** (0.033)						
ENESM					0.268*** (0.101)					
MV				-0.000*** (0.000)	0.000*** (0.000)					
MV ²				0.000*** (0.000)	-0.000*** (0.000)					
RANK				0.000* (0.000)	-0.000*** (0.000)					
INST				-0.603*** (0.061)	0.177*** (0.062)					
IV1						0.323*** (0.009)				
IV2								0.300*** (0.007)		
Treat×Post										0.034*** (0.011)

^⑤ In the reverse causality test, all control variables are lagged by one period. To conserve space, these lagged control variables retain their original variable names in the presentation.

LEV	-0.085*** (0.029)	-0.587*** (0.180)	-0.015 (0.036)	-0.488*** (0.070)	0.122** (0.053)	-0.186* (0.099)	-0.084*** (0.022)	-0.134 (0.090)	-0.084*** (0.022)	-0.085*** (0.026)
PAY	0.016** (0.007)	0.025 (0.048)	0.004 (0.008)	0.070*** (0.017)	0.016** (0.008)	0.019 (0.023)	0.015*** (0.006)	-0.005 (0.023)	0.015*** (0.006)	0.024*** (0.006)
ROA	-0.043 (0.055)	-0.546 (0.382)	-0.014 (0.058)	-0.793*** (0.249)	0.181* (0.094)	-0.245 (0.221)	-0.041 (0.049)	-0.477** (0.226)	-0.041 (0.049)	-0.064 (0.053)
BOARD	-0.009 (0.022)	0.097 (0.137)	0.022 (0.029)	-0.117** (0.059)	0.060*** (0.017)	0.032 (0.082)	-0.009 (0.018)	0.074 (0.081)	-0.009 (0.018)	-0.002 (0.020)
DUAL	-0.005 (0.008)	0.039 (0.046)	-0.003 (0.009)	0.045* (0.023)	-0.017** (0.007)	-0.023 (0.025)	-0.005 (0.006)	-0.018 (0.024)	-0.005 (0.006)	-0.005 (0.007)
EPS	0.009 (0.006)	0.122*** (0.040)	0.002 (0.007)	-0.003 (0.020)	0.015*** (0.004)	0.062*** (0.021)	0.009 (0.006)	0.077*** (0.020)	0.009 (0.006)	0.010* (0.006)
GROWTH	-0.009** (0.004)	-0.023 (0.028)	0.003 (0.006)	0.018 (0.024)	-0.010* (0.005)	-0.015 (0.016)	-0.009** (0.004)	-0.001 (0.016)	-0.009** (0.004)	-0.008** (0.004)
MB	-0.052*** (0.018)	0.195* (0.108)	-0.018 (0.021)	0.114 (0.077)	-0.038** (0.019)	0.323*** (0.060)	-0.054*** (0.014)	0.272*** (0.058)	-0.055*** (0.014)	-0.044*** (0.016)
TOP5	0.053 (0.043)	-0.683*** (0.231)	0.059 (0.058)	-0.327*** (0.084)	0.121*** (0.038)	-0.019 (0.120)	0.054* (0.031)	-0.160 (0.113)	0.054* (0.031)	0.049 (0.037)
SIZE	0.038*** (0.008)	0.164*** (0.045)	0.012 (0.011)	0.129*** (0.037)	0.000 (0.016)	0.046* (0.026)	0.038*** (0.006)	0.001 (0.024)	0.038*** (0.006)	0.034*** (0.007)
Constant	2.014*** (0.177)	-3.086*** (1.127)		-2.721*** (0.732)	2.447*** (0.336)	-1.150** (0.573)		0.168 (0.539)		1.955*** (0.164)
Year	√	√	√	√	√	√	√	√	√	√
Firm	√	√	√			√	√	√	√	√
Industry				√	√					
Obs	7,628	5,771	4,337	7,628	7,628	7,628	7,592	7,628	7,592	9,938
AR(1) P			0.000							
AR(2) P			0.834							
Hansen P			0.288							
K P LM							782.527***		788.044***	
K P Wald F							2676.450		2877.474	
10% maximal IV size							16.38		16.38	
R2_a	0.696	0.114		0.153	0.607	0.551	0.637	0.571	0.637	0.732

Notes:***p < 0.01; **p < 0.05; *p < 0.1. t statistics are in parentheses

4.6 Heterogeneity analysis

4.6.1 Institutional Investor

Institutional investors have the ability to supervise listed companies(Song and Song,2020)^[55].The more institutional investors hold shares in a company, enterprises are more proactive in responding to negative emotions through ESG disclosure. Meanwhile, institutional investors can be divided into two categories: long-term and short-term. Long-term institutional investors focus more on the sustainable development of enterprises, while short-term

institutional investors are more concerned with short-term returns. Referring to research of Zhou et al. (2017)^[56], we conducts grouped tests based on the median proportion of long-term institutional shareholding ratios. According to the results of columns (1) to (2) of Table 8, we find that in firms with higher long-term institutional shareholding ratios, the coefficient of ENESM is more significant. This indicates that long-term institutional investors pay greater attention to corporate sustainable development and can exert a more substantial external governance effect on firms.

4.6.2 Internal Control

Both internal control mechanisms and external media can function as effective monitoring and complementary governance mechanisms for corporations. We conducted group testing based on the median of Dibo's internal control index, and the results are shown in columns (3) to (4) of Table 8. Compared to samples of low internal control quality, the coefficient of ENESM is more significant in samples with high internal control quality. The findings convincingly demonstrate that when a firm's internal control quality is weak, external media can fully play its supervisory role and motivate the firm to enhance its EDQ.

TABLE 8 Results of heterogeneity tests on institutional investors and internal control

	EDQ	EDQ	EDQ	EDQ
	Low long-term institutional shareholding ratio	High long-term institutional shareholding ratio	Low internal control quality	High internal control quality
	(1)	(2)	(3)	(4)
ENESM	0.004 (0.003)	0.007* (0.004)	-0.001 (0.004)	0.011*** (0.003)
LEV	-0.088** (0.035)	-0.069 (0.045)	-0.138*** (0.047)	-0.003 (0.036)
PAY	0.010 (0.009)	0.016 (0.010)	-0.001 (0.011)	0.028*** (0.009)
ROA	-0.086 (0.065)	0.060 (0.092)	0.034 (0.151)	-0.040 (0.060)
BOARD	-0.001 (0.030)	-0.003 (0.036)	0.018 (0.039)	-0.024 (0.028)
DUAL	0.009 (0.010)	-0.017 (0.011)	0.006 (0.013)	-0.009 (0.010)
EPS	0.015* (0.009)	0.005 (0.009)	0.012 (0.010)	0.002 (0.009)
GROWTH	-0.002 (0.006)	-0.009 (0.007)	-0.014** (0.007)	-0.002 (0.007)
MB	0.019 (0.024)	-0.081*** (0.026)	-0.097*** (0.029)	-0.025 (0.024)
TOP5	0.101* (0.054)	0.031 (0.066)	-0.047 (0.063)	0.179*** (0.054)
SIZE	0.021** (0.010)	0.041*** (0.014)	0.048*** (0.013)	0.017* (0.009)

constant	2.384*** (0.228)	1.931*** (0.311)	2.053*** (0.291)	2.258*** (0.214)
Year	√	√	√	√
Firm	√	√	√	√
Obs	3,865	3,763	3,765	3,765
R2_a	0.687	0.704	0.708	0.677

Notes:***p < 0.01; **p < 0.05; *p < 0.1. t statistics are in parentheses

4.7 Further study

4.7.1 ESG Disclosure Preference

We have confirmed that ENESM can encourage companies to improve the EDQ. However, ESG information covers three aspects: E (environment), S (society), and G (governance). Considering the costs and benefits of information disclosure, companies may not be able to simultaneously improve the disclosure quality in E, S, and G. To this end, we used the values of E, S, and G from Bloomberg's ESG database to examine whether companies have disclosure preferences when facing negative reviews. The results are shown in Table 9. Compared to E and S, the coefficient of ENESM on G is more significant, indicating that companies prefer to improve the G information quality.

This may be attributed to the governance (G) dimension's comparative advantages in terms of transformation costs, controllability, and signaling efficiency. Improvements in G primarily rely on institutional revisions and structural optimizations (e.g., establishing specialized committees, refining executive compensation mechanisms). Such reforms fall entirely within the decision-making scope of the board of directors and are characterized by low implementation resistance, short execution cycles, and minimal capital investment. In contrast, substantive enhancements in environmental (E) and social (S) information often require costly technological upgrades and extended timeframes, making it difficult to rapidly improve EDQ. Consequently, when facing public pressure, companies tend to prioritize rapidly enhancing G information quality to restore reputation.

T A B L E 9 Regression results of ESG information disclosure preference

	E (1)	S (2)	G (3)
ENESM	0.015 (0.015)	-0.004 (0.006)	0.006*** (0.002)
LEV	-0.093 (0.168)	-0.113 (0.076)	-0.059** (0.025)
PAY	0.072* (0.042)	0.002 (0.019)	0.010 (0.006)
ROA	0.321 (0.369)	0.058 (0.147)	-0.058 (0.045)
BOARD	0.013 (0.130)	-0.052 (0.061)	0.009 (0.019)
DUAL	-0.044 (0.048)	-0.021 (0.021)	0.001 (0.006)
EPS	0.020 (0.040)	0.005 (0.015)	0.004 (0.004)

GROWTH	-0.091*** (0.028)	-0.033*** (0.012)	0.002 (0.004)
MB	-0.491*** (0.109)	-0.162*** (0.045)	0.030** (0.014)
TOP5	-0.113 (0.227)	-0.088 (0.103)	0.060* (0.036)
SIZE	0.265*** (0.044)	0.169*** (0.023)	-0.013* (0.007)
constant	-5.884*** (0.962)	-1.746*** (0.534)	4.030*** (0.154)
Year	√	√	√
Firm	√	√	√
Obs	7,628	7,628	7,628
R2_a	0.371	0.285	0.695

Notes:***p < 0.01; **p < 0.05; *p < 0.1. t statistics are in parentheses

4.7.2 ESG Greenwashing

In China, there is a lack of unified standards and effective supervision for the disclosure of ESG information by enterprises. In order to cater to the preferences of rating agencies, meet the requirements of green financing conditions and corporate environmental image, companies have the motivation to implement ESG greenwashing(EGW) (Huang, 2022) ^[57]. Referring to the measurement of EGW by Yu et al. (2022) ^[58], we use Bloomberg ESG score to measure ESG disclosure, and use ESG score from the CNRDS database to measure the actual ESG performance. The difference between the standardized results of the above two is used to measure EGW. In order to verify the EGW motivation of enterprises, we divided the sample into three samples(high, medium, and low EDQ samples) to conduct grouped regression tests for Model (1). The results are shown in columns (1) to (3) of Table 10. Compared to samples with low EDQ and high EDQ, the coefficient of ENESM is more significant in samples with medium EDQ. The above results indicate that when EDQ is medium, firms are more likely to respond to ENESM through EGW.

So, will EGW be subject to regulatory constraints? Existing research has demonstrated that corporate greenwashing practices can be effectively mitigated under public scrutiny(Zhou and Chen,2024) ^[59].We argue heavy polluting enterprises face stricter ESG disclosure requirements.Heavily polluting enterprises face broader and deeper environmental and social responsibility inspections. Heavily polluting enterprises will be more inclined to substantially improve the EDQ under regulatory constraints. Therefore, we conduct grouping tests based on the heavy polluting industries. As shown in columns (4) and (5) of Table 10, the coefficient of ENESM in the samples of non heavy polluting industry enterprises is significantly positive, fully indicating that the behavior of enterprises responding to ENESM through EGW can be effectively constrained under strong regulation.

Moreover, there are similarities between the measurement of ESG rating divergence(ESGRD) and EGW in the existing literature. The measurement of EGW is typically based on the standardized difference between two types of ESG scores, while ESGRD is often measured using the standard deviation of scores from multiple ESG rating agencies. To demonstrate the robustness of the EGW findings in our further analysis, we replaced the EGW variable with ESGRD and re-ran the regression. Referring to Avramov et al. (2022)^[60], ESGRD is calculated based on the ESG scores from six rating agencies: Bloomberg, Wind, SinoSec, FTSE Russell, SynTao Green Finance, and Menglang. The calculation procedure involves: first converting all ratings into numerical scores, then ranking and standardizing

these scores, and finally computing the average of the pairwise standard deviations of the rankings across all agencies. As shown in columns (6) to (8) of Table 10, regardless of the level of EDQ, the impact of ENESM on ESGRD is statistically insignificant. This indicates that while ENESM can lead to EGW, its effect on ESGRD is relatively weak.

T A B L E 10 Regression results of ESG greenwashing motivation

	EGW	EGW	EGW	EGW	EGW	ESGRD	ESGRD	ESGRD
	Low EDQ	Medium EDQ	Hign EDQ	Heavy polluting enterprises	Non heavy polluting enterprises	Low EDQ	Medium EDQ	Hign EDQ
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ENESM	0.006 (0.238)	0.043* (1.777)	0.015 (0.608)	0.029 (1.248)	0.038** (1.976)	-0.002 (0.004)	-0.004 (0.003)	0.003 (0.004)
LEV	-0.658*** (-2.589)	-0.105 (-0.417)	-0.681** (-2.191)	-0.199 (-0.793)	-0.798*** (-3.916)	-0.046 (0.044)	-0.026 (0.041)	0.174*** (0.047)
PAY	0.083 (1.112)	-0.065 (-1.049)	-0.026 (-0.355)	0.045 (0.730)	-0.008 (-0.161)	0.032** (0.013)	0.001 (0.010)	0.008 (0.010)
ROA	-0.247 (-0.407)	0.657 (1.219)	0.706 (1.068)	0.750 (1.453)	-0.398 (-0.899)	0.005 (0.096)	-0.130* (0.073)	0.115 (0.087)
BOARD	0.187 (0.753)	-0.004 (-0.015)	0.212 (0.841)	0.231 (1.013)	-0.185 (-1.064)	-0.006 (0.040)	-0.014 (0.033)	0.040 (0.033)
DUAL	-0.079 (-1.042)	0.073 (0.914)	0.105 (1.442)	0.057 (0.813)	0.054 (0.885)	-0.006 (0.010)	0.008 (0.008)	-0.007 (0.011)
EPS	0.034 (0.415)	-0.000 (-0.004)	-0.040 (-0.768)	0.049 (0.953)	0.018 (0.354)	-0.017 (0.011)	0.017 (0.010)	-0.004 (0.007)
GROWTH	0.061 (1.291)	-0.034 (-0.808)	-0.066 (-1.288)	-0.052 (-1.240)	0.022 (0.665)	-0.004 (0.006)	0.009 (0.006)	-0.008 (0.006)
MB	0.269 (1.327)	-0.150 (-0.992)	0.103 (0.589)	0.026 (0.164)	-0.368*** (-2.866)	-0.003 (0.031)	-0.003 (0.023)	0.011 (0.024)
TOP5	-0.227 (-0.608)	0.199 (0.525)	-0.139 (-0.262)	0.047 (0.130)	0.178 (0.602)	-0.076 (0.059)	0.057 (0.061)	-0.004 (0.064)
SIZE	0.013 (0.158)	-0.035 (-0.538)	0.288*** (3.174)	0.051 (0.648)	0.109** (2.256)	0.010 (0.012)	-0.002 (0.011)	-0.032** (0.013)
Constant	-2.378 (-1.395)	2.137 (1.276)	-5.429*** (-2.657)	-3.337* (-1.842)	-1.797 (-1.635)	-0.433* (0.250)	0.298 (0.292)	0.853*** (0.290)
Year	√	√	√	√	√	√	√	√
Firm	√	√	√	√	√	√	√	√
Obs	2,545	2,542	2,541	2,998	4,630	2,525	2,532	2,536
R2_a	0.158	0.0373	0.0587	0.304	0.299	0.00956	0.0295	0.143

Notes:***p < 0.01; **p < 0.05; *p < 0.1. t statistics are in parentheses

4.7.3 The impact of non-ESG negative emotions on social media

The baseline regression examines the compelling effect of ENESM on the EDQ. However, do non-ESG

social media emotions also influence EDQ? To address this question, this study tests the impact of non-ESG negative sentiment (NENESM) on EDQ. The result is presented in column (1) of Table 11, revealing a positive correlation between the two. This finding does not negate the conclusions of the main analysis but instead sheds light on the complex motivations underlying corporate responses to negative emotions. Drawing on reputation insurance theory, this study interprets this phenomenon as follows: when a firm encounters NENESM, its overall reputation is impaired, prompting the firm to signal its continued role as a responsible entity through higher EDQ as a means of reputation repair. This suggests that ESG disclosure serves not only as a remedial tool in response to ENESM but also as a general instrument of impression management. This finding is not contradictory to the main effect, it enriches our understanding of firms' strategic motives for disclosure.

4.7.4 The internal logic behind analysts' attention

The preceding analysis has demonstrated that ENESM leads to a reduction in analyst attention (AA), which in turn imposes pressures such as reputational damage and tightened financing channels on firms, indirectly motivating them to improve EDQ. However, beyond the quantitative change in analyst behavior reflected by the decline in AA, what deeper aspects of analyst behavior trigger market pressures on firms? This study posits that analyst forecast dispersion (DISP) and forecast error (FE) further capture the market pressures that firms may face. When ENESM occurs, even analysts who continue to follow the firm may exhibit forecast divergence and misjudgment due to ambiguous signals, thereby transmitting risk signals about the firm's uncertain prospects to the capital market and amplifying corporate pressure. Therefore, building on the analysis of the AA mechanism, further examining the transmission pathways of analyst forecast dispersion (DISP) and forecast error (FE) can further clarify the underlying mechanism of analyst attention. To this end, this study measures DISP and FE following the approach of Chu et al. (2019)^[61] and tests whether ENESM compels firms to improve EDQ by exacerbating DISP and FE. The empirical results are presented in columns (2) and (3) of Table 11, indicating that ENESM indeed increases both DISP and FE.

TABLE 11 Regression results of non-ESG negative emotions consequences and the analyst attention path

VARIABLES	EDQ (1)	DISP (2)	FE (3)
NENESM	0.004** (0.002)		
ENESM		0.000** (0.000)	0.001** (0.000)
LEV	-0.082*** (0.029)	0.002 (0.002)	-0.007 (0.005)
PAY	0.015** (0.007)	0.001 (0.000)	0.001 (0.001)
ROA	-0.040 (0.055)	-0.027*** (0.008)	-0.153*** (0.017)
BOARD	-0.009 (0.022)	-0.002 (0.002)	-0.007* (0.004)
DUAL	-0.005 (0.008)	-0.001 (0.000)	-0.001 (0.001)
EPS	0.009	-0.001**	-0.013***

	(0.006)	(0.001)	(0.002)
GROWTH	-0.009**	-0.000	-0.003***
	(0.004)	(0.000)	(0.001)
MB	-0.053***	0.002*	-0.005**
	(0.018)	(0.001)	(0.003)
TOP5	0.054	-0.001	-0.006
	(0.043)	(0.002)	(0.006)
SIZE	0.039***	0.000	0.003***
	(0.008)	(0.000)	(0.001)
Year	√	√	√
Firm	√	√	√
Constant	1.997***	-0.007	-0.040
	(0.178)	(0.012)	(0.028)
Obs	7,624	5,630	6,193
R2_a	0.695	0.0659	0.335

Notes:***p < 0.01; **p < 0.05; *p < 0.1. t statistics are in parentheses

4.7.6 Threshold test of “cost inversion”

The preceding analysis has examined firms’ disclosure strategies based on the logic of “cost inversion”. Specifically, when the cost of maintaining a low level of EDQ (implicit cost) exceeds the cost of improving EDQ (explicit cost), firms’ disclosure behavior reverses, leading them to invest more in enhancing EDQ. As the mechanism of “cost inversion” has already been tested from the perspectives of analyst attention, financing costs, and customer stability, we now conduct an empirical test of the “cost inversion” mechanism at the firm level. We argue that the “cost inversion” scenario occurs when firms face financial constraints or existential crises. This study posits that the “cost inversion” scenario occurs when firms face financial constraints or existential crises. Therefore, we select financing constraint and bankruptcy risk as threshold variables to examine how firms enhance EDQ based on “cost inversion” when facing ENESM.

We measure financing constraint using the KZ index and bankruptcy risk using Oscore. The threshold test results are presented in Table 12. The threshold value for KZ is 0.256 (P = 0.025), and the threshold value for Oscore is -10.818 (P = 0.060). The regression results are shown in Table 13^⑥. The KZ analysis indicates that when KZ is below the threshold value (i.e., when firms have ample financial resources), the cost reversal effect of ENESM has not yet been triggered, and the coefficient is not significant. When KZ exceeds the threshold value (i.e., when firms face financial constraints), the pressure of implicit costs forces firms to take action, and the effect of ENESM on EDQ improvement becomes significantly positive. The Oscore analysis shows that when Oscore is below the threshold value (i.e., when firms have low bankruptcy risk), the coefficient of ENESM is significantly negative. This suggests that among financially healthy firms, management’s motivation to save disclosure costs dominates. However, when bankruptcy risk crosses the threshold, the coefficient becomes significantly positive. This indicates that for firms on the verge of distress, the implicit costs triggered by ENESM far outweigh the explicit costs of improving disclosure, compelling them to enhance EDQ to avoid greater

⑥The DUAL and year fixed effect are omitted from the regression due to collinearity. Therefore, in the threshold regression model of this paper, we remove the DUAL and year fixed effect, and instead incorporate a Year variable (a time trend variable) to account for the influence of time.

economic losses.

T A B L E 12 Result of Threshold value test on KZ and Oscore

Threshold variable	Threshold value	P value	F value	Critical value		
				1%	5%	10%
KZ	0.2556	0.025	12.52**	16.311	10.187	8.794
Oscore	-10.8183	0.040	11.17*	14.998	10.785	9.145

T A B L E 13 Regression results of threshold model test on KZ and Oscore

VARIABLES	EDQ (1)	EDQ (2)
ENESM	-0.006	
(KZ<0.2556)	(0.004)	
ENESM	0.006**	
(KZ>0.2556)	(0.003)	
ENESM		-0.016***
(Oscore<-10.8183)		(0.005)
ENESM		0.006**
(Oscore>-10.8183)		(0.003)
LEV	-0.159***	-0.156***
	(0.026)	(0.027)
PAY	0.004	0.004
	(0.007)	(0.007)
ROA	-0.028	-0.004
	(0.064)	(0.065)
BOARD	-0.027	-0.025
	(0.021)	(0.021)
EPS	0.006	0.004
	(0.006)	(0.006)
GROWTH	0.004	0.003
	(0.005)	(0.005)
MB	-0.073***	-0.078***
	(0.015)	(0.015)
TOP5	0.046	0.027
	(0.033)	(0.033)
SIZE	0.040***	0.041***

	(0.006)	(0.006)
Year	0.061***	0.061***
	(0.001)	(0.001)
Firm	√	√
Constant	-120.040***	-120.436***
	(2.226)	(2.236)
Obs	5,010	4,900
R2_a	0.584	0.585

Notes:***p < 0.01; **p < 0.05; *p < 0.1. t statistics are in parentheses

5. CONCLUSIONS AND POLICY IMPLICATIONS

5.1 Conclusions

With the development of technology, expressing opinions through social media will become the norm for investors, and enterprises need to pay long-term attention to the expression of investors' opinions. This paper conducts research from the perspective of ESG disclosure, taking A-share listed companies from 2011 to 2021 as the research objects to examine the impact of ENESM on the EDQ. The research findings are as follows: Firstly, ENESM will force enterprises to improve the EDQ. After conducting robustness tests such as replacing variable measurement methods, changing regression models, adopting Propensity Score Matching (PSM) and Two-Stage Least Squares Regression (2SLS), the conclusion still holds. Secondly, the mechanism analysis reveals that ENESM leads to decreased AA, increased FC, and weakened CS, which in turn forces enterprises to improve the EDQ. It is further confirmed through a two-stage model that enterprises' improvement in the EDQ can hedge against the above adverse effects caused by ENESM. Thirdly, the heterogeneity test shows that the promoting effect of ENESM on EDQ becomes stronger when there is a higher proportion of long-term institutional investors' shareholding and lower internal control quality. Fourthly, further research finds that compared with E (Environment) information disclosure and S (Social) information, ENESM have a stronger effect on improving the G (Governance) information quality. Meanwhile, although ENESM lead to companies with medium EDQ turning ESG greenwashing, strong regulation can effectively constrain this behavior. Not only ENESM, but non-ESG negative emotions also compels companies to improve EDQ. ENESM further leads to analyst forecast dispersion and error. Moreover, increased financing constraints and bankruptcy risk compel companies to invest more in ESG disclosure costs in response to ENESM. Through an in-depth exploration of the AA mechanism, the study finds that ENESM further leads to analyst forecast dispersion and error. Moreover, companies can weigh the costs and benefits of disclosing ESG information, increased financing constraints and bankruptcy risk compel companies to invest more in ESG disclosure costs in response to ENESM.

5.2 Policy implications

Based on the above conclusions, this paper puts forward the following suggestions:

First, enterprises should reasonably utilize ESG disclosure for public opinion management. On the one hand, firms need to pay attention to the spread of ENESM and be alert to its potential adverse effects, such as reduced AA, increased FC, and decreased CS. This study finds that higher EDQ can effectively mitigate the negative impacts of ENESM. Enterprises should strengthen their awareness of the costs and benefits of ESG disclosure, treating

investments in ESG disclosure as strategic inputs to increase AA, reduce FC, and secure CS. They should also formulate appropriate ESG disclosure strategies based on their financial conditions and profitability to alleviate the pressures brought by ENESM in a timely manner. On the other hand, firms should enhance their awareness of ESG disclosure by increasing the frequency, channels, and forms of such disclosure. For example, they can release ESG reports semi-annually or quarterly, actively utilize diverse social media platforms such as WeChat and investor conferences to promote their ESG image, and adopt charts, short videos, and other formats to translate professional content into easily understandable information. Through these measures, enterprises can strengthen their positive image in the eyes of stakeholders such as customers and investors. Furthermore, firms need to increase their sensitivity to ENESM, acknowledge the continuously improving expertise and governance demands of investors, and rely on robust internal control systems as a foundation for disclosure. By effectively leveraging information exchange on social media, they can substantially improve ESG disclosure quality as a response.

Secondly, social media platforms should effectively play the roles of external supervision and information interaction. They should continuously optimize the "voice-expressing" mode of investors, create a clean and orderly online communication space, and provide effective channels for small and medium-sized investors to "put forward suggestions". They can set up ESG-related theme forums to strengthen the supervision intensity and scope of investors on enterprises' ESG disclosure. In addition, social media platforms can also add ESG-related stock recommendation sections. Through big data, they can count the mutual relationship between the quality of enterprises' ESG disclosure and their market value, popularize the importance of ESG, and indirectly guide enterprises to improve the EDQ.

Thirdly, government departments should attach importance to the ESG promotion effect of social media. They can set up official accounts on social media platforms to popularize ESG-related content among the platform's audience and achieve an overall improvement in ESG awareness. They can also continuously pay attention to negative emotions about enterprises on the platforms, promptly feed back the opinions of shareholders to the target enterprises, and guide the target enterprises to improve the EDQ in response. In addition, government departments can monitor the response speed of enterprises' ESG disclosure under ENESM, and implement certain rewards for enterprises with a faster improvement speed and higher quality, so as to encourage enterprises to actively respond to ENESM by improving the EDQ.

5.3 Research limitations and prospects

Although this paper examines the relationship between ENESM and EDQ and clarifies their influencing mechanisms, there are still some deficiencies in data and research design. First, case studies can be conducted by focusing on specific enterprises. Second, the propagation effect of ENESM on EDQ within supply chains should also be considered.

Data Availability Statement: The data supporting this study have been provided as a supplementary attachment.

References:

[1]Wang D, Sun K P&Gao H(2020).The Impact of"Voting with Your Mouth"on Management Voluntary Disclosure. *Journal of Financial Research*, (11): 188–206.

- [2]Li H.Y,L Q.Y,Li S.Y&Fu S.Y. Environmental, Social and Governance Information Disclosure and Enterprise Green Innovation Performance. *Statistical Research*, 2022, 39(12): 38-54.
- [3]Li J C, Ren G Q,ZhaoM J&Wang M D(2025). Investor ESG Information Dissemination and Corporate ESG Performance: Empirical Evidence Based on Online Interactive Platform. *Journal of Finance and Economics*, 251(5): 34-49.
- [4]Dyck A, Volchkova N&Zingales L(2008).The Corporate Governance Role of the Media: Evidence from Russia.The *Journal of Finance*, 63: 1093-1135.
- [5]Lee L F,Hutton A P&Shu S(2015). The Role of Social Media in the Capital Market: Evidence from Consumer Product Recalls. *Journal of Accounting Research*,53(2): 367-404.
- [6]Yang F&Zhang Y.M(2021). "Investors' 'Voicing' on Social Media and Enterprise Technological Innovation——Research Based on Information Effect and Governance Effect". *Journal of Shanxi University of Finance and Economics*,43(11): 115-126.
- [7]Luo J.B&Dou C(2022). Does the Social Media "Voicing" of Minority Shareholders Affect the Merger and Acquisition Preferences of Enterprises? ——Empirical Evidence Based on the Guba (e Interaction) Platform. *Journal of Shanghai University of Finance and Economics*,24(04): 123-137.
- [8]Wang D,Sun K.P&G H(2020).The Impact of "Voting with the Mouth" on Social Media on Management's Voluntary Earnings Forecasts. *Journal of Financial Research*,(11): 188-206.
- [9]Xie J,Nozawa W,Yagi M,Fujii H&Managi S(2019).Do environmental, social, and governance activities improve corporate financial performance?Business Strategy and the Environment,28(2): 286-300.
- [10]Reber B, Gold A& Gold S(2022). ESG disclosure and idiosyncratic risk in Initial Public Offerings. *Journal of Business Ethics*,179(3): 867-886.
- [11]Reza H. Chowdhury, Fu C B, Huang Q P & Lin N Y (2021). CSR disclosure of foreign versus U.S. firms: Evidence from ADRs.*Journal of International Financial Markets, Institutions and Money*, (70),101275.
- [12]Danjue Clancey-Shang & C B Fu(2024).CSR disclosure, political risk and market quality: Evidence from the Russia-Ukraine conflict.*Global Finance Journal*,(60),100938.
- [13]Bagh T,Bouri E&Khan M A (2025), Climate change sentiment, ESG practices and firm value: international insights. *China Finance Review International*, 15(3):608–635
- [14]Dai Z F & Jiang Q N (2025).Climate policy uncertainty and corporate ESG performance: evidence from Chinese listed companies.*China Finance Review International*,15(3):578-607.
- [15]Qureshi M A, Kirkerud S, Theresa K&Ahsan T(2020) The impact of sustainability (environmental, social, and governance) disclosure and board diversity on firm value: The moderating role of industry sensitivity. *Business Strategy and the Environment*,29 (3) : 1199-1214.
- [16]Zhou F Z, Pan W Y,& Fu H (2020)The ESG Responsibility Performance of Listed Companies and the Shareholding Preferences of Institutional Investors——Empirical Evidence from A-share Listed Companies in China . *Scientific Decision Making*, (11): 15-41
- [17]Hermalin B E,& Weisbach M S(2012)Information Disclosure and Corporate Governance. *The Journal of Finance*, 67(1): 195–233.
- [18]Lin W L , Law S H , Ho J A&Sambasivan M(2019)The causality direction of the corporate social responsibility-Corporate financial performance Nexus: Application of Panel Vector Autoregression approach.The *North American Journal of Economics and Finance*, 48(APR.):401-418.
- [19]Wang S J, T Y, &Dang L L(2022)The Fulfillment of ESG Responsibilities, Competitive Strategies and Financial Performance of Industrial Enterprises . *Accounting Research*, (3): 77-92.
- [20]Dou C, &Luo J B(2020)Can Minority Shareholders' "Voicing" on Social Media Improve Executive Compensation Contracts? *Finance & Trade Economics*, 41(12): 85-100.
- [21]Wu P, Zhang J, Gu Y Y& Chen Y X(2022)Research on the Impact of Negative Media Reports on the Disclosure of Key Audit Matters——Taking Sina Weibo as an Example. *Audit & Economy Research*, 37(5): 33-42.
- [22]Liu X X, &Li M H(2018)Media Negative Reports, Analyst Tracking and Tax Aggressiveness . *Accounting Research*, (9): 64-71.
- [23]Dellavigna S , &Pollet J M (2009)Investor Inattention and Friday Earnings Announcements.The *Journal of Finance*, 64(2):709-749.
- [24]Zhang Z X, &Wu Z Y(2021)Media Sentiment Contagion and Analysts' Optimistic Bias——Empirical Evidence Based on Machine Learning Text Analysis Method . *Management World*, 37(01): 170-185 + 11 + 20 – 22.
- [25]EFRAG. 2021. ESRG 1 Double Materiality Conceptual Guidelines for Standard-setting (Working Paper) , 21-31.
- [26] Zheng J D, Lü X L, Lü B& G F (2022) Information Interaction on Social Media Platforms and Capital Market Pricing Efficiency——Evidence Based on Hundreds of Millions of Big Data from the Guba Forum. *The Journal of Quantitative & Technical Economics*, 39(11): 91-112.
- [27]Asante-Appiah B,& Lambert T A (2022)The role of the external auditor in managing environmental, social, and governance (ESG) reputation risk. *Review of Accounting Studies*,27(8):1-53.

- [28] La R F & Bernini F. (2022). ESG controversies and the cost of equity capital of European listed companies: the moderating effects of ESG performance and market securities regulation. *International Journal of Accounting & Information Management*, 30(5):641-663.
- [29] He Q, Zhuang P T, Xia Q, & Ju W J. (2025). How Does ESG News Sentiment Affect Corporate Inefficient Investment Behavior? A Dual Perspective of Information and Sentiment. *Journal of Financial Research*, (11):96-114.
- [30] Zeng Z & Tang S (2023) The Role of State-owned Enterprises as Economic Stabilizers under the COVID-19 Pandemic—From the Perspective of Supply Chain Support. *Economic Research Journal*, 58(3):78–96.
- [31] Xiao H J, Shen H T, & Zhou Y K (2024) Customer Digitalization, Supplier ESG Performance and Supply Chain Sustainability. *Economic Research Journal*, 59(3):54-73.
- [32] Freeman R E (1984) *Strategic Management: A Stakeholder Approach*. Boston: Pitman Publishing Inc: 24-25.
- [33] Raimo N, Caragnano A, Zito M, Vitolla F & Mariani M (2021) Extending the benefits of ESG disclosure: The effect on the cost of debt financing. *Corporate Social Responsibility and Environmental Management*, 28(4): 1412-1421.
- [34] Tan X. (2016). Research on the Governance Function of Analyst Attention—Based on the Examination of Two Types of Agency Costs. *Securities Market Herald*, (12):37-45.
- [35] Yu Z B, Tian G L, Qi B L & Zhang H (2011) The Corporate Governance Mechanism of Media Attention—Based on the Examination from the Perspective of Earnings Management. *Management World*, (09):127-140.
- [36] Li Y, Wu Y C, & Tian X Y (2023) Enterprise ESG Performance and Supply Chain Discourse Power. *Journal of Finance and Economics*, 49(08):153-168.
- [37] Barry C B and Brown S J (1984) Differential Information and the Small Firm Effect. *Journal of Financial Economics*, 13(2):283—295.
- [38] Handa P and Linn S (1993) Arbitrage Pricing with Estimation Risk. *Journal of Financial and Quantitative Analysis*, 28(1):81—100.
- [39] Dahiya M, & Singh S (2021) The linkage between CSR and cost of equity: an Indian perspective. *Sustainability Accounting, Management and Policy Journal*, 12(3), 499-521.
- [40] Li Z B, Shao Y M, Li Z Z, & Li M S (2022) ESG Information Disclosure, Media Supervision and Corporate Financing Constraints [J]. *Scientific Decision Making*, (7): 1-26.
- [41] Orlitzky M, Schmidt, F L, & Rynes, S L (2003) Corporate social and financial performance: A meta-analysis. *Organization studies*, 24(3), 403-441.
- [42] Atif M, & Ali S (2021) Environmental, social and governance disclosure and default risk. *Business Strategy and the Environment*, 30(8), 3937-3959.
- [43] Li S, Li Y Y & Du Y G (2023) Customer Stability and Firm's Cost of Equity Capital—Analysis Based on Risk Taking Perspective. *Systems Engineering*, 41(2): 72-88.
- [44] Huang C & Jiang Q S (2022) The Impact of Guba Activity on Stock Liquidity. *Chinese Journal of Management* 19(11), 1714-1724.
- [45] Zhang J R, Song X Y, & Ying X Q (2025) How Does the Online Voices of Minority Shareholders Affect Corporate Forward-looking Information Disclosure? — Evidence from Investor Interactive Platform. *Systems Engineering—Theory & Practice*, 1-29.
- [46] Cai G L, Zhang Y N, Xu Y, & Lu R (2022) Investor-Listed Company Interaction and the Resource Allocation Efficiency in the Capital Market: Evidence Based on the Cost of Equity. *Journal of Management World*, 38(08), 199-217.
- [47] Pan H B & Zhang H (2020) Executive-Customer Relationship and Customer Stability. *Chinese Journal of Management*, 17(02):196-203.
- [48] Kim Y, Su L N, Wang Z, and Wu H (2021) The Effect of Trade Secrets Law on Stock Price Synchronicity: Evidence from the Inevitable Disclosure Doctrine. *Accounting Review*, 96(1), 325—348.
- [49] Luo J H & Du X Q (2014), Media Coverage, Institutional Environment and Stock Price Crash Risk. *Accounting Research*, (09): 53-59+97.
- [50] Hou W & Ding R (2015) Retail Investor Attention and Stock Liquidity. *Journal of International Financial Markets Institutions and Money*, 37:12-26.
- [51] Zhang X W, Zhang T T & C F Cao (2022) Enterprise Nonlocal Development under Tax Sharing. *Journal of Finance and Economics*, 48(06):140-154.

- [52] An Z, Chen C, Naiker V & Wang J(2020)Does media coverage deter firms from withholding bad news? Evidence from stock price crash risk. *Journal of Corporate Finance*, 64: 101664.
- [53] Lewbel A (1997) Constructing Instruments for Regressions With Measurement Error When no Additional Data are Available, with an Application to Patents and R&D. *Econometrica*, 65(5): 1201-1213.
- [54] Xi M M, LI T & L Q Y Zhang(2025) “More Words Than Deeds”in Corporate Environmental Responsibility Behavior under Online Media Supervision: A Classified AI Measurement Using Large Language Models. *Journal of Lanzhou University (Social Sciences)*, 53(3): 159-176.
- [55] Song Y L , Song Y H (2020) Institutional Investors' Shareholdings and Accounting Information Quality from the Perspective of Certified Public Accountants—Empirical Evidence from Audit Adjustments. *Accounting Research*, (11): 136-151.
- [56] Zhou S N, Zhang Q S & L X Hu(2017).Can Institutional Investors Promote M&A Performance of State - owned Enterprises?—Discussion on Institutional Investors Heterogeneity in China. *Accounting Research*, (06): 67-74+97.
- [57] Huang S Z(2022)The 'greenwashing' and anti greenwashing in ESG reports. *Finance and Accounting Monthly*, (1): 3-11.
- [58] Ellen Pei-Yi Yu & Bac Van Luu(2020) Catherine Huirong Chen. Greenwashing in environmental, social and governance disclosures. *Research in International Business and Finance*, 52: 101192.
- [59] Zhou Y & Chen L (2024) "Public attention and “Environmental Disclosure Greenwashing” : pressure from oversight or incentive from legitimacy". *China Finance Review International*, Vol. ahead-of-print No. ahead-of-print.
- [60] Avramov D, Cheng S, Lioui A & Tarelli A(2022) Sustainable investing with ESG rating uncertainty . *Journal of financial economics*, 145(2) : 642-664.
- [61] Chu Jian, Qin Xuan and Fang Junxiong(2019)Margin-trading, Short-selling and Analysts' Forecast Optimism. *Journal of Management World*,(1):151-228.